

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Medium

Question

Prices of 14 Different Cars

Type of car	Priced at no more than \$25,000	Priced greater than \$25,000	Total
Nonhybrid	5	3	8
Hybrid	2	4	6
Total	7	7	14

The table above shows information about 14 cars listed for sale on an auto dealership’s website. If one of the cars listed for sale is selected at random, what is the probability that the car selected will be a hybrid car priced at no more than \$25,000 ?

Answer

A. $\frac{1}{7}$

B. $\frac{2}{7}$

C. $\frac{1}{3}$

D. $\frac{4}{7}$

Correct Answer: A

Rationale

Choice A is correct. It’s given that there are 2 hybrid cars priced at no more than \$25,000. It’s also given that there are 14 cars total for sale. Therefore, the probability of selecting a hybrid priced at no more than \$25,000 when one car is chosen at random is $\frac{2}{14} = \frac{1}{7}$.

Choice B is incorrect. This is the probability of selecting a hybrid car priced greater than \$25,000 when choosing one car at random. Choice C is incorrect. This is the probability, when choosing randomly from only the hybrid cars, of selecting one priced at no more than \$25,000. Choice D is incorrect. This is the probability of selecting a hybrid car when selecting at random from only the cars priced greater than \$25,000.